

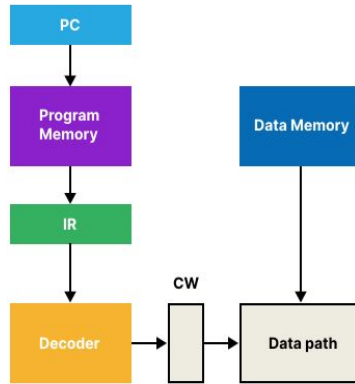
Device aka Всякая всячина про железо

Марк Блуменау

CISC vs RISC



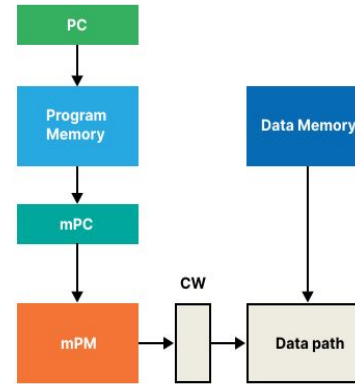
RISC



Simple instructions
One instruction=One CW PM longer



CISC



Complex instructions
One instruction = Several CW
PM Standard

А кто есть CUDA?

Сложно.

Через N лет может эта лекция устареет

AMD Athlon K7/K8

В принципе набор инструкций x86-64

Apple A4/A5

AMD Zen

Tesla AI

И сейчас TensTorrent



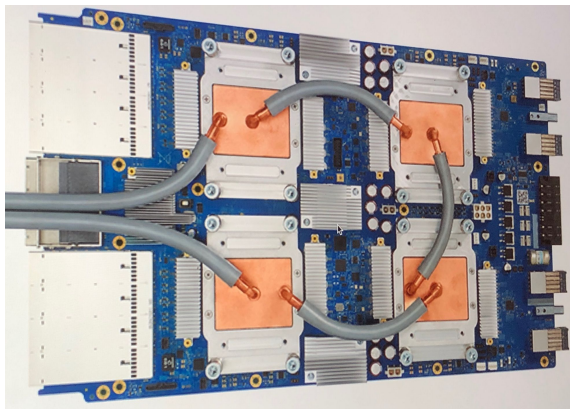
Jim Keller

Specification	p100a	p150a	p150b
Part Number	TC-03008	TC-03003	TC-03002
Tensix Cores	120	140	140
AI Clock	1.35 GHz	1.35 GHz	1.35 GHz
“Big RISC-V” Cores	16	16	16
SRAM	180 MB	210 MB	210 MB
Memory	28 GB GDDR6	32 GB GDDR6	32 GB GDDR6
Memory Speed	16 GT/sec	16 GT/sec	16 GT/sec
Memory Bandwidth	448 GB/sec	512 GB/sec	512 GB/sec
TeraFLOPS (FP8)	664	774	774
TeraFLOPS (FP16)	166	194	194
TeraFLOPS (BLOCKFP8)	332	387	387
TBP (Total Board Power)	300W	300W	300W
External Power	12V-2x6	12V-2x6	12V-2x6
Connectivity	N/A	4x QSFP-DD 800G (Passive)*	4x QSFP-DD 800G (Passive)*
System Interface	PCI Express 5.0 x16	PCI Express 5.0 x16	PCI Express 5.0 x16
Cooling	Active	Active	Passive
Dimensions (WxDxH)	42mm x 270mm x 111mm	42mm x 270mm x 111mm	42mm x 270mm x 111mm

А какие девайсы вообще есть в торче?

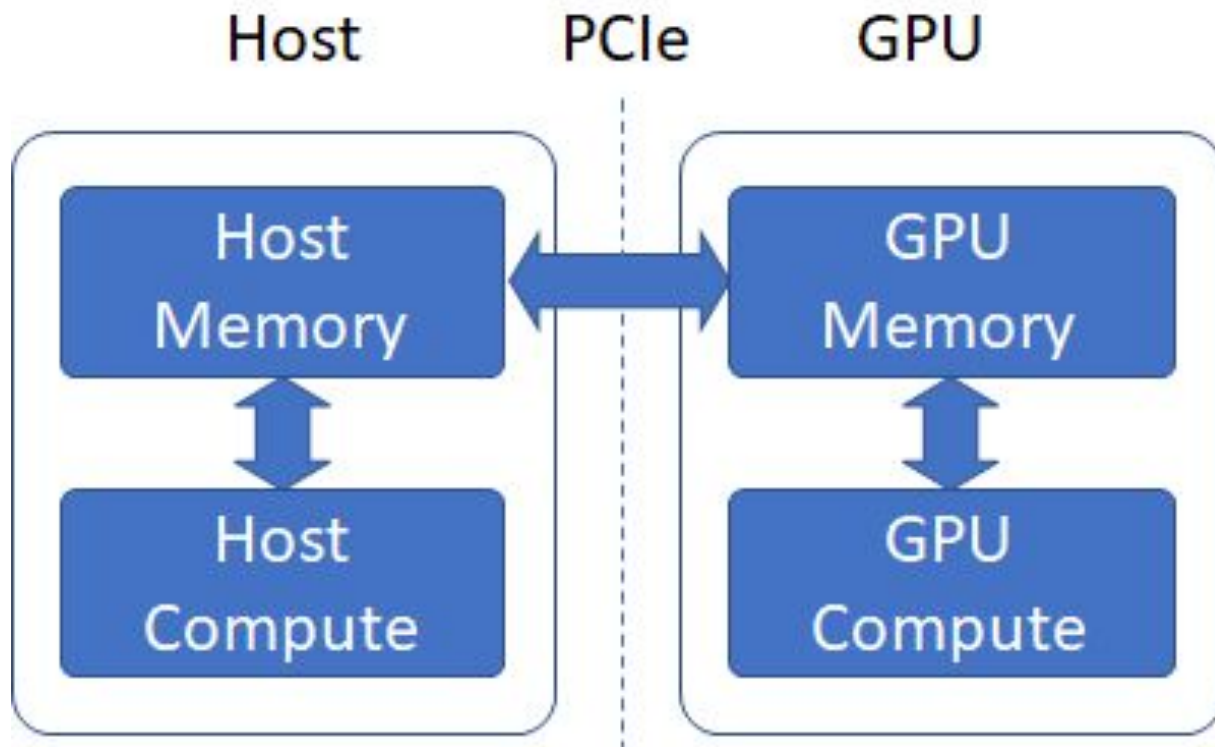


С костылями



intel[®]
ARC[™]

Как работает компьютер?

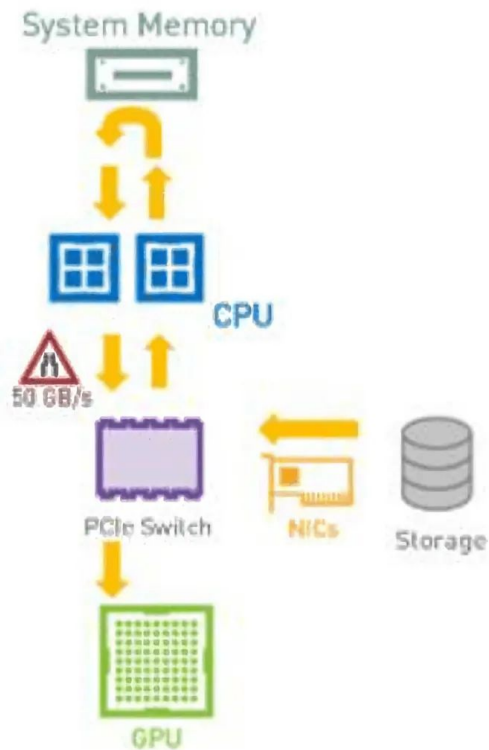


Насколько PCi-e быстрый?

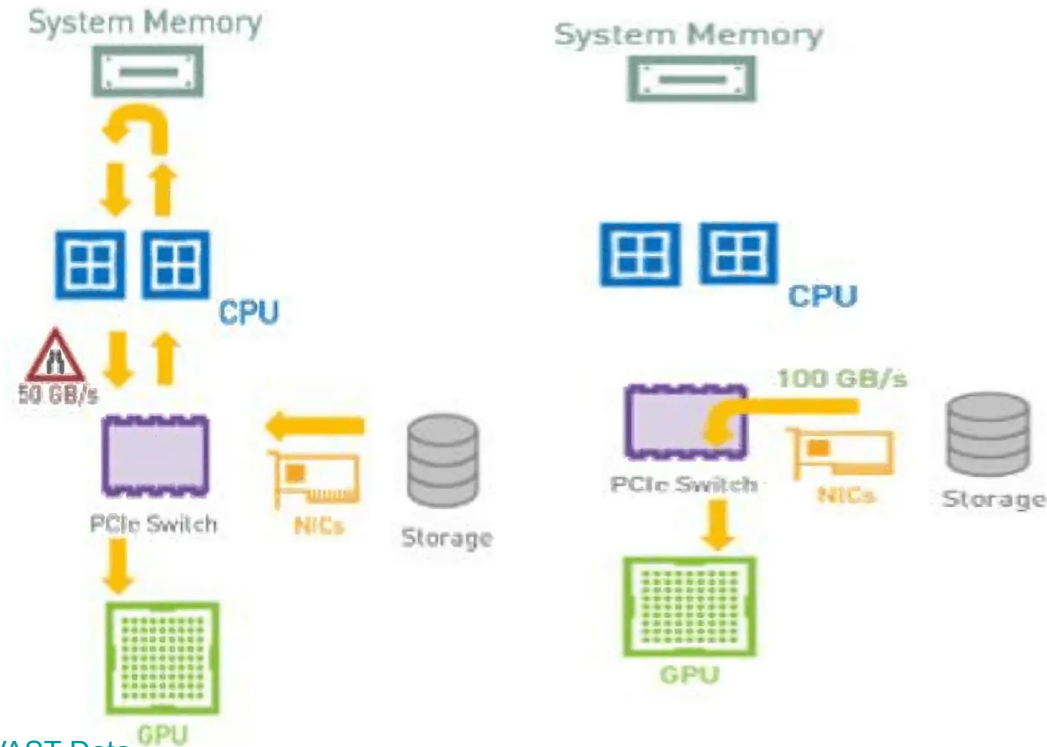
PCI Express link performance^{[57][58]}

Version	Intro-duced	Line code		Transfer rate (per lane) ^{[i][ii]}	Throughput ^{[i][iii]}				
					x1	x2	x4	x8	x16
1.0	2003	NRZ	8b/10b	2.5 GT/s	0.250 GB/s	0.500 GB/s	1.000 GB/s	2.000 GB/s	4.000 GB/s
2.0	2007			5.0 GT/s	0.500 GB/s	1.000 GB/s	2.000 GB/s	4.000 GB/s	8.000 GB/s
3.0	2010		128b/130b	8.0 GT/s	0.985 GB/s	1.969 GB/s	3.938 GB/s	7.877 GB/s	15.754 GB/s
4.0	2017			16.0 GT/s	1.969 GB/s	3.938 GB/s	7.877 GB/s	15.754 GB/s	31.508 GB/s
5.0	2019			32.0 GT/s	3.938 GB/s	7.877 GB/s	15.754 GB/s	31.508 GB/s	63.015 GB/s
6.0	2022	PAM-4 FEC	1b/1b 242B/256B FLIT	64.0 GT/s 32.0 GBd	7.563 GB/s	15.125 GB/s	30.250 GB/s	60.500 GB/s	121.000 GB/s
7.0	2025 (planned)			128.0 GT/s 64.0 GBd	15.125 GB/s	30.250 GB/s	60.500 GB/s	121.000 GB/s	242.000 GB/s

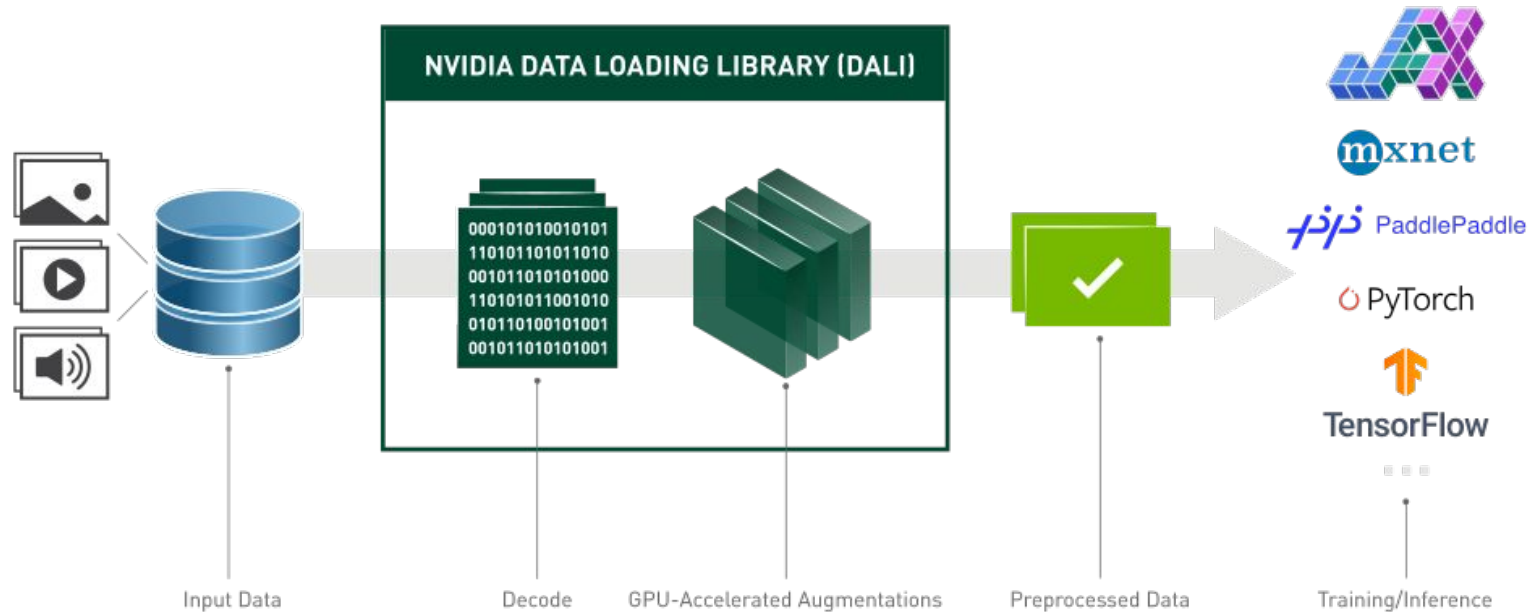
А как у нас текут данные?



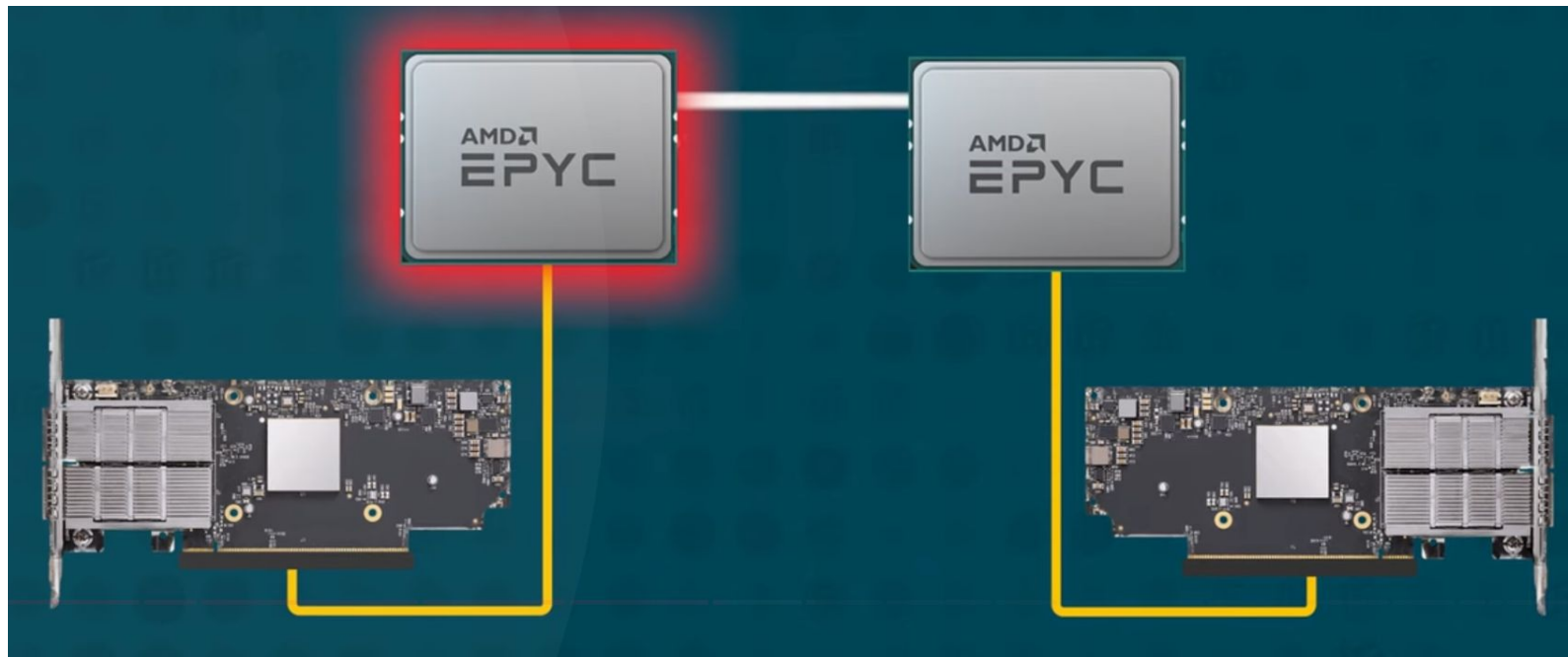
А можно ли лучше? (GPU Direct Storage)



DALI

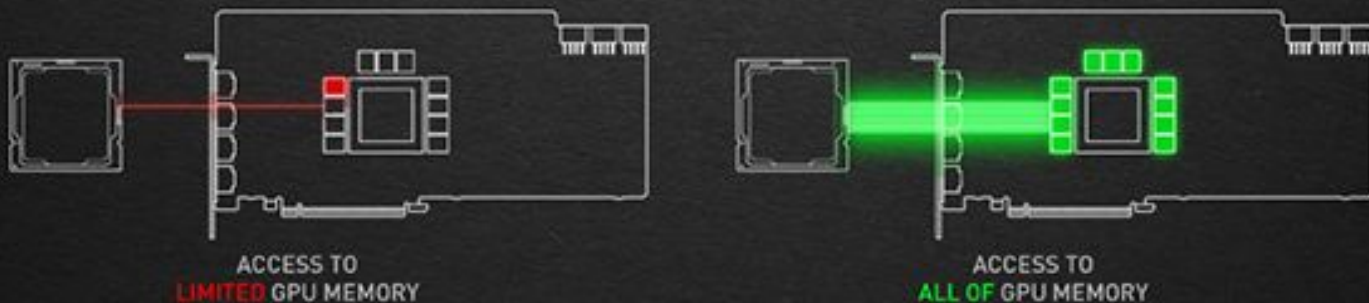


Проблемы кластеров

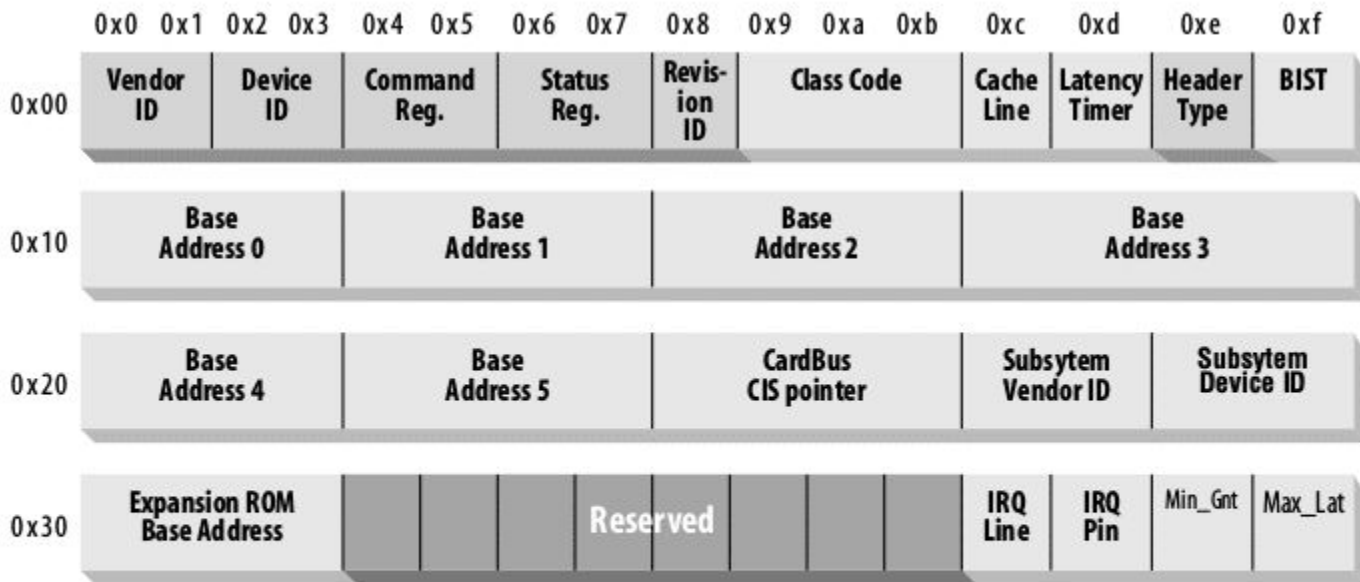


Продолжаем перевод с геймерского

NVIDIA Resizable BAR Technology



Base Address Register (или 256 Мб, адреса не 64-бита)



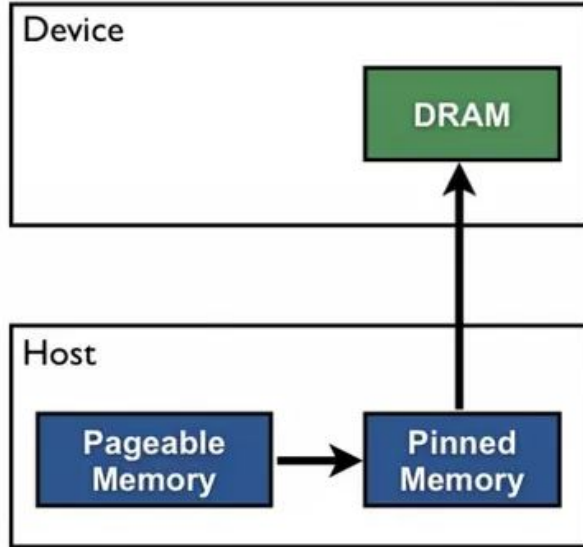
 - Required Register

 - Optional Register

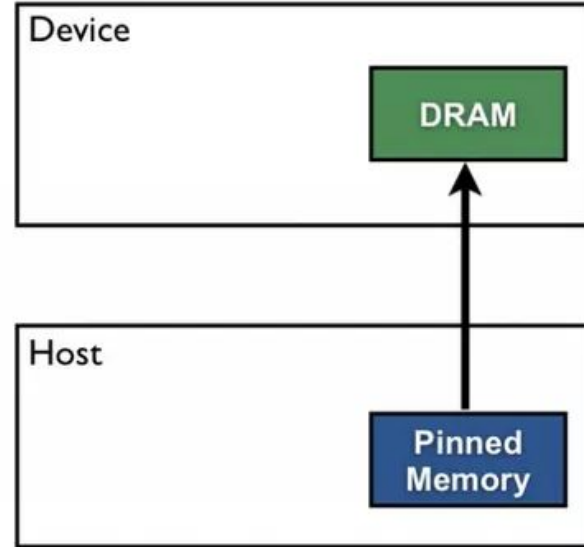
[What is the Base Address Register \(BAR\) in PCIe?](#)

Pinned memory

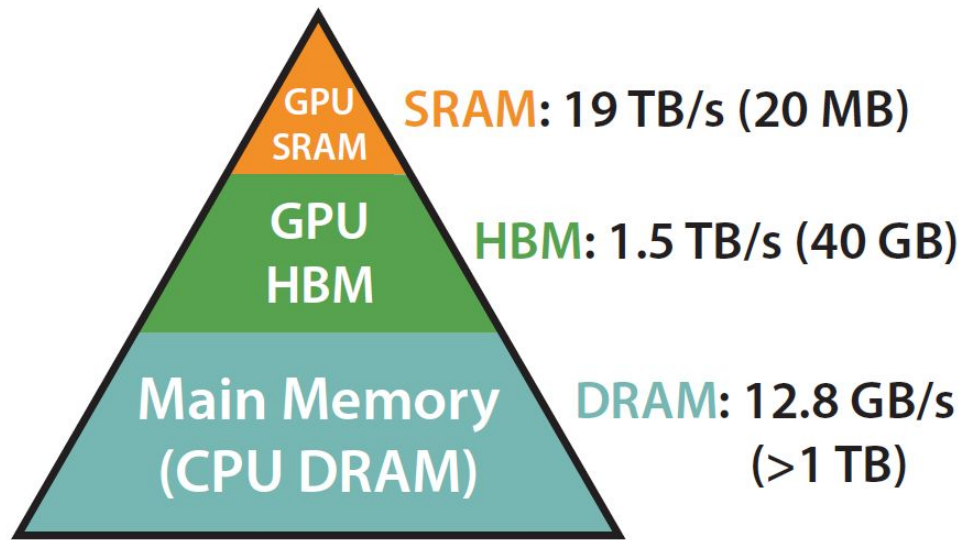
Pageable Data Transfer



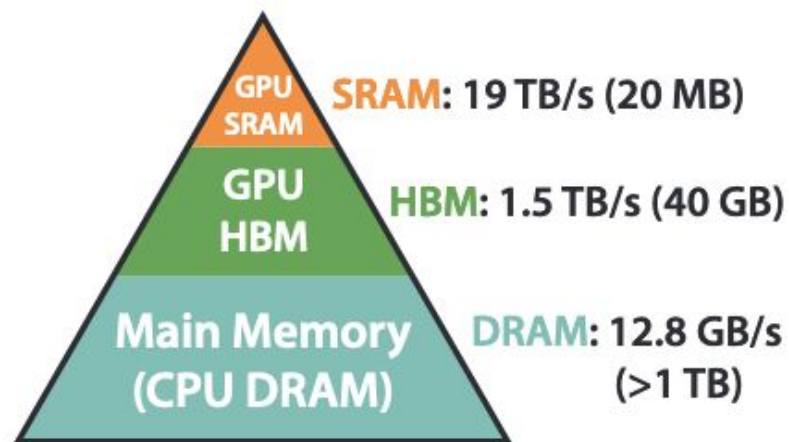
Pinned Data Transfer



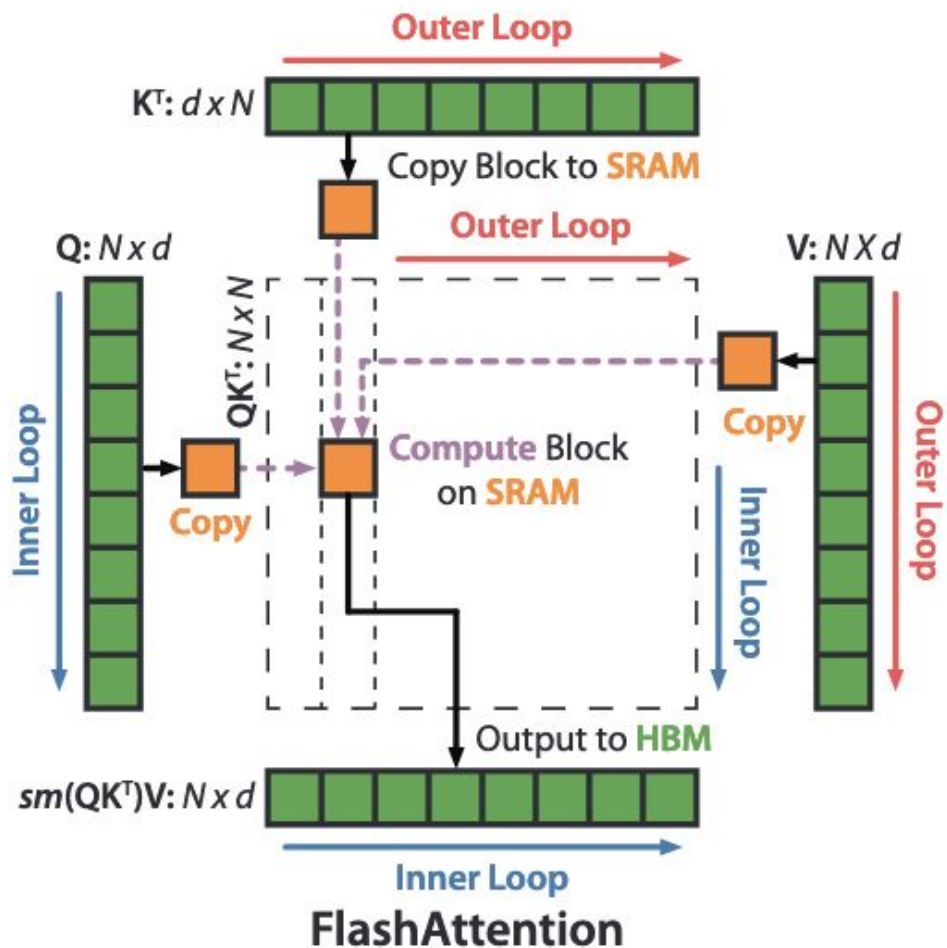
На чём живет Flash Attention и немного Layer Fusing?



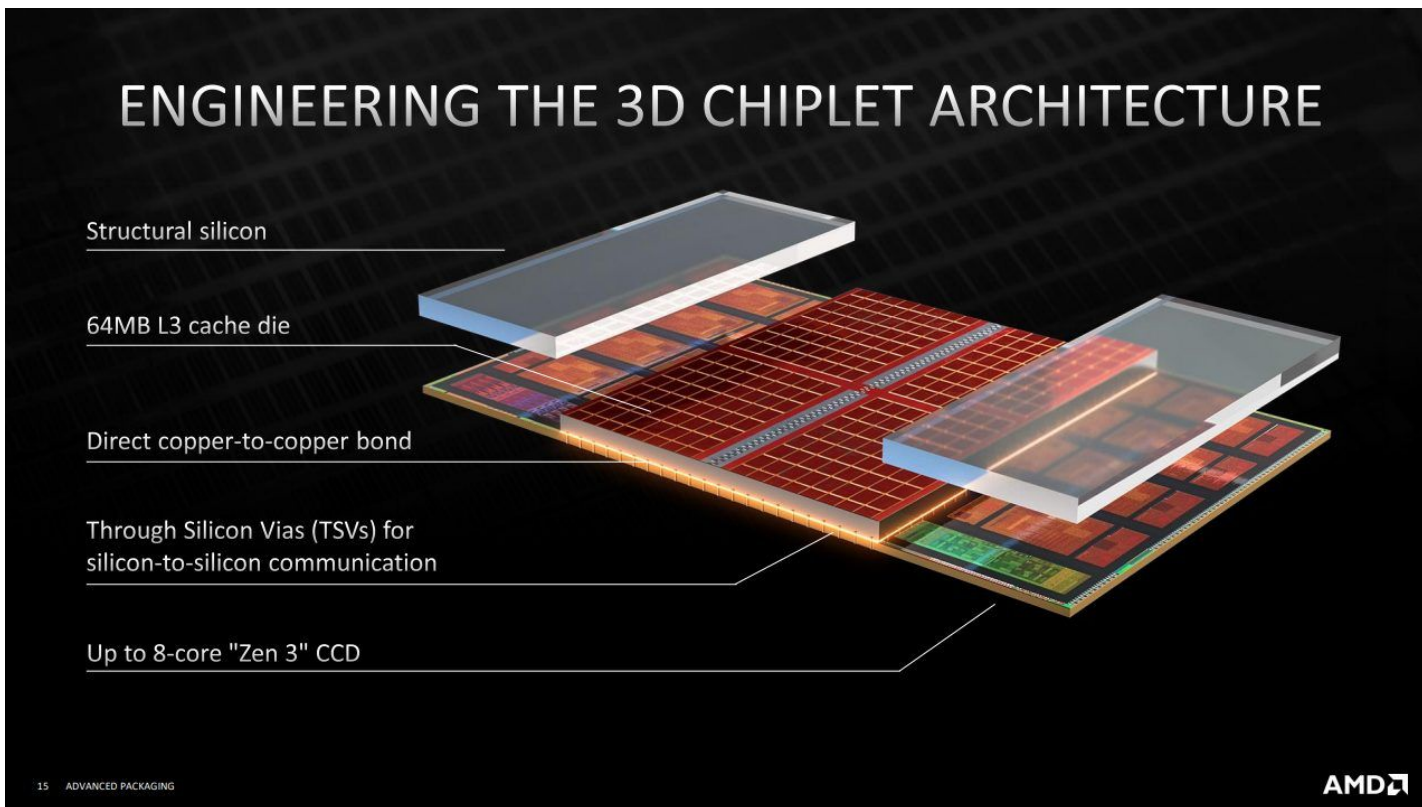
Memory Hierarchy with
Bandwidth & Memory Size



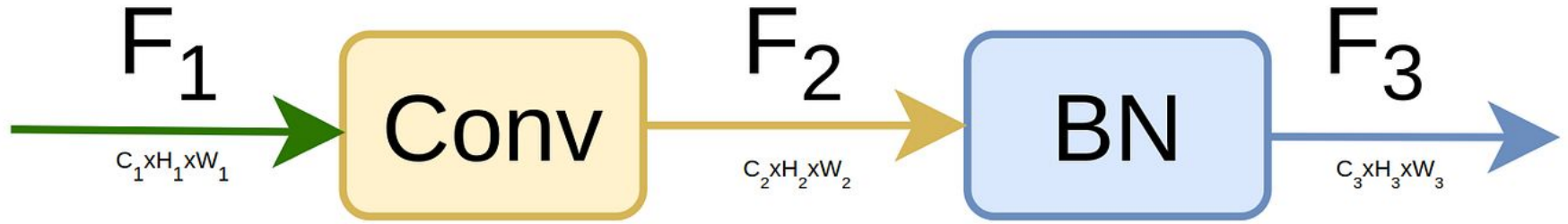
Memory Hierarchy with Bandwidth & Memory Size



Очередной перевод с геймерского



Layer fusion



Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\dots m}\}$;
Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$

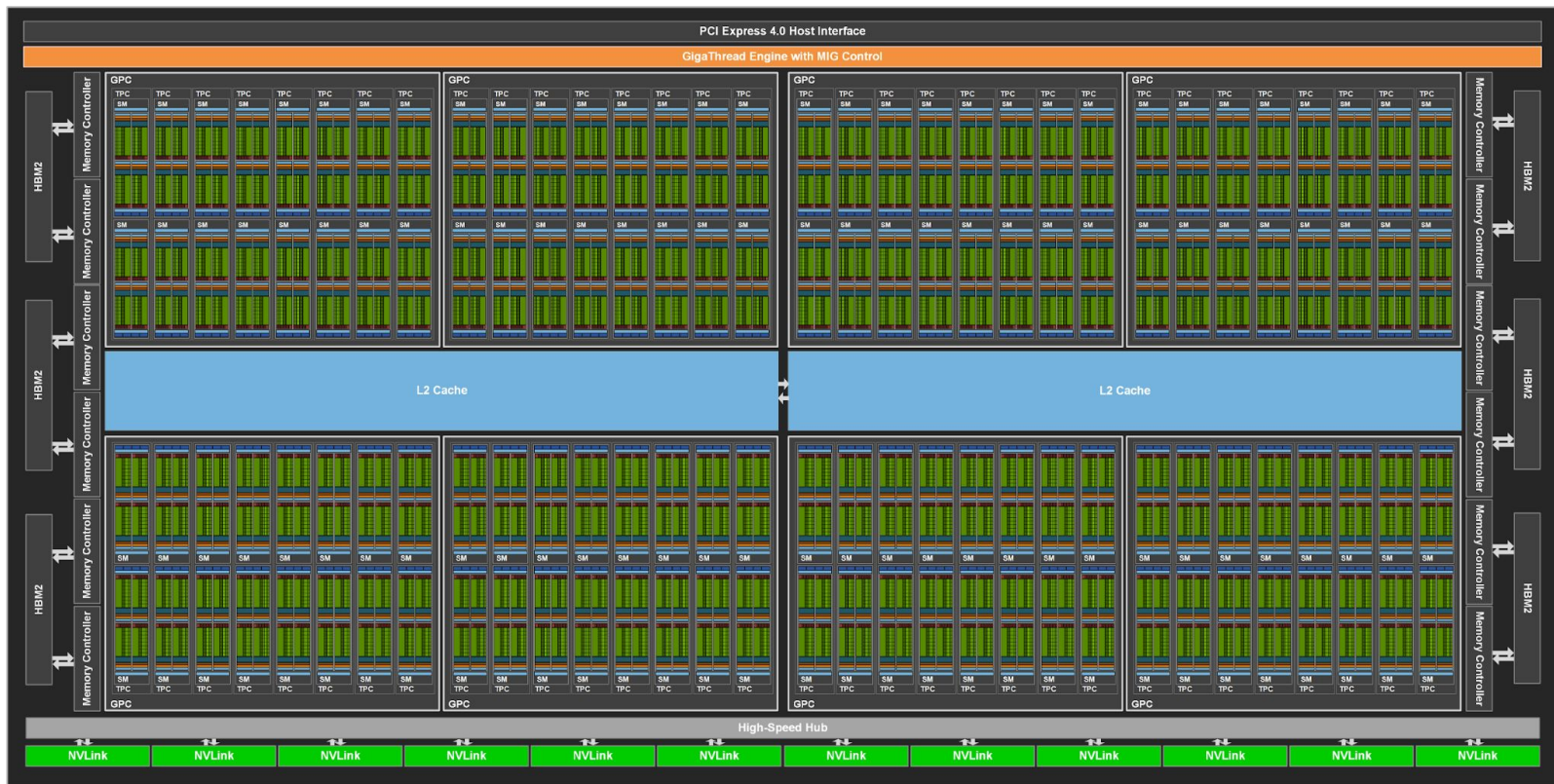
$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$

Результат

$$W' = \gamma \frac{W}{\sqrt{\sigma^2 + \epsilon}}$$

$$b' = \beta - \frac{\gamma \mu}{\sqrt{\sigma^2 + \epsilon}}$$

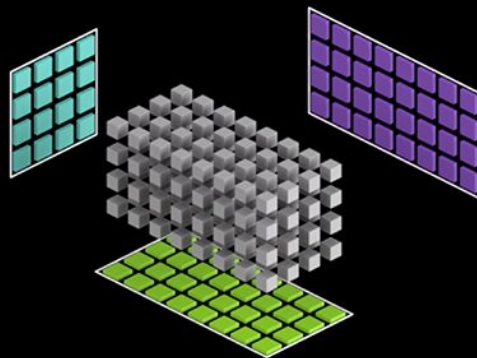
Как выглядит видеокарта изнутри?



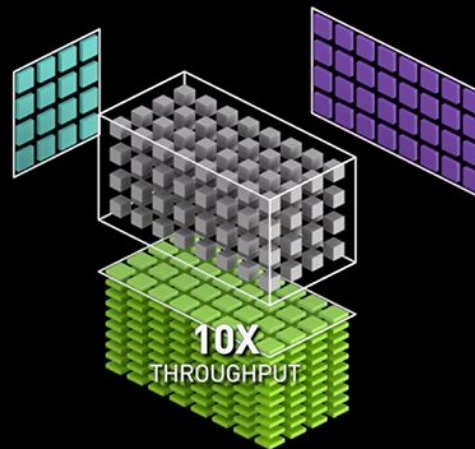


А что за тензорные ядра?

NVIDIA V100 FP32



NVIDIA A100 Tensor Core TF32



Ну вы поняли...



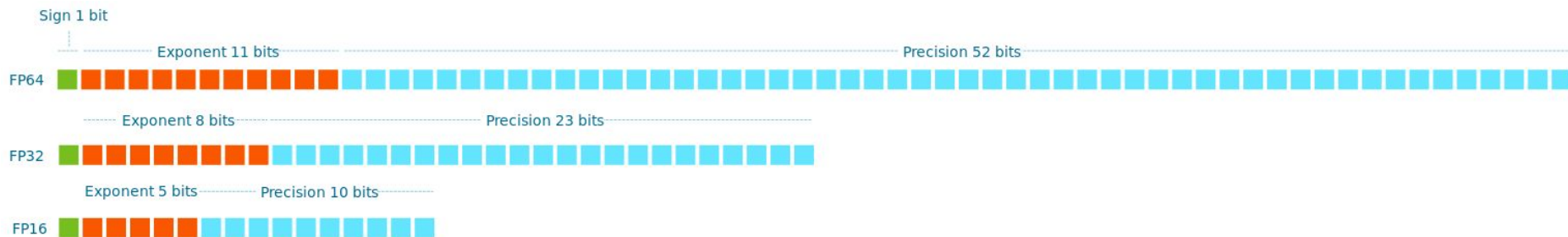
[Amazon.com: ASUS NVIDIA GeForce RTX 5090 Video Card 32GB GDDR7 PCI Express 5.0 / ROG-ASTRAL-RTX5090-O32G-GAMING : Electronics](https://www.amazon.com/ASUS-NVIDIA-GeForce-RTX-5090-Video-Card-32GB-GDDR7-PCI-Express-5.0/dp/B0F7H8H8H8)

И зачем нам эти ядра?

	A100 40GB PCIe	A100 80GB PCIe	A100 40GB SXM	A100 80GB SXM
FP64	9.7 TFLOPS			
FP64 Tensor Core	19.5 TFLOPS			
FP32	19.5 TFLOPS			
Tensor Float 32 (TF32)	156 TFLOPS 312 TFLOPS*			
BFLOAT16 Tensor Core	312 TFLOPS 624 TFLOPS*			
FP16 Tensor Core	312 TFLOPS 624 TFLOPS*			
INT8 Tensor Core	624 TOPS 1248 TOPS*			
GPU Memory	40GB HBM2	80GB HBM2e	40GB HBM2	80GB HBM2e
GPU Memory Bandwidth	1,555GB/s	1,935GB/s	1,555GB/s	2,039GB/s
Max Thermal Design Power (TDP)	250W	300W	400W	400W
Multi-Instance GPU	Up to 7 MIGs @ 5GB	Up to 7 MIGs @ 10GB	Up to 7 MIGs @ 5GB	Up to 7 MIGs @ 10GB
Form Factor	PCIe		SXM	

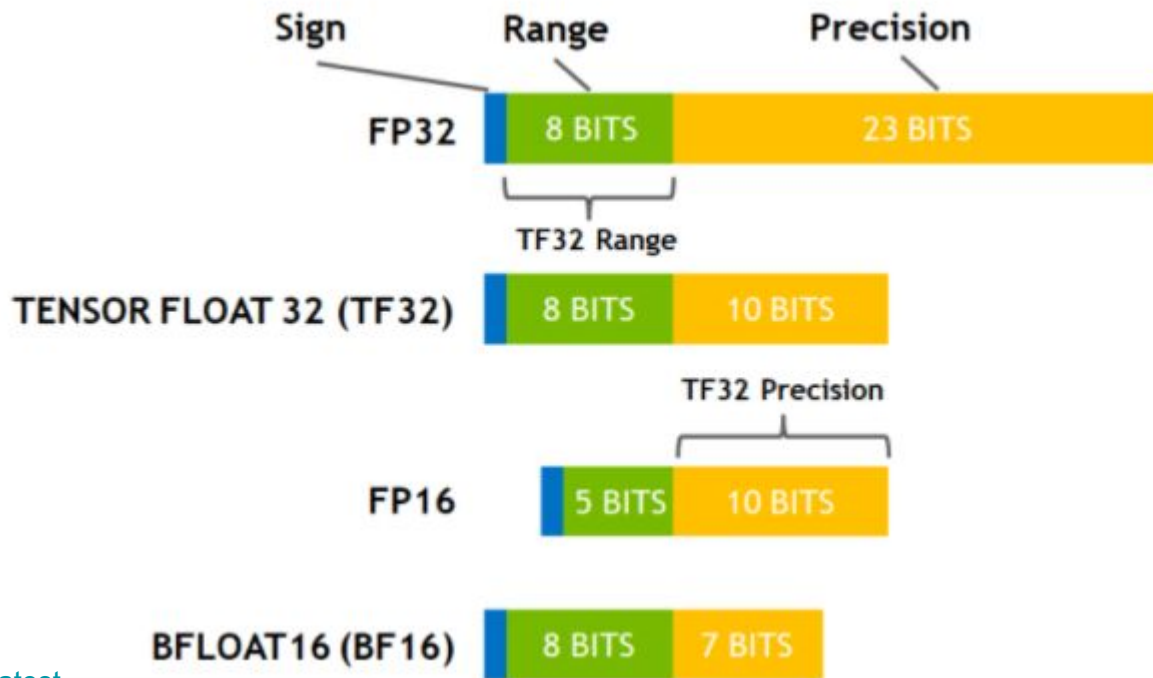
<https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/a100/pdf/nvidia-a100-datasheet-us-nvidia-1758950-r4-web.pdf>

Давайте поиграем!



Вопрос: сколько бит в TF32?

Или коротко о том, что названия в железах перестали нести смысл

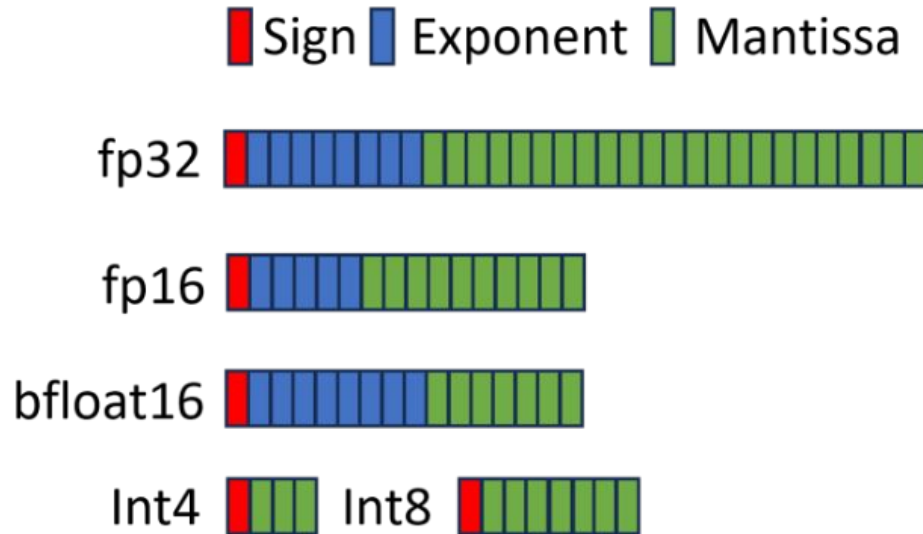


А понятно ли зачем они теперь?

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Form Factor	PCIe		SXM	

<https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/a100/pdf/nvidia-a100-datasheet-us-nvidia-1758950-r4-web.pdf>

INT 4, INT 8



Стоп, стоп, стоп, а как????



Unsloth

<https://unsloth.ai> › blog › d...



Run DeepSeek-R1 Dynamic 1.58-bit

27 янв. 2025 г. — The 1.58bit quantization should fit in 160GB of VRAM for fast inference (2x H100 80GB), with it attaining around 140 tokens per second for ...

<https://unsloth.ai/blog/deepseekr1-dynamic>

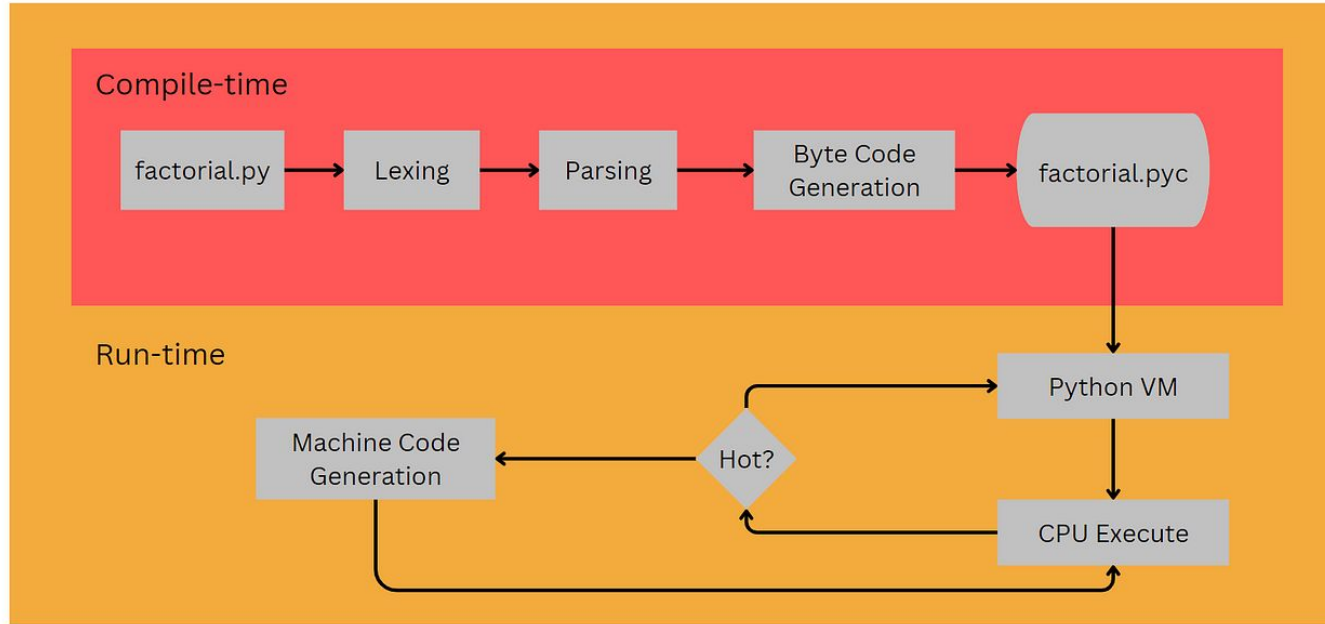
:)

$\log_2(3) =$

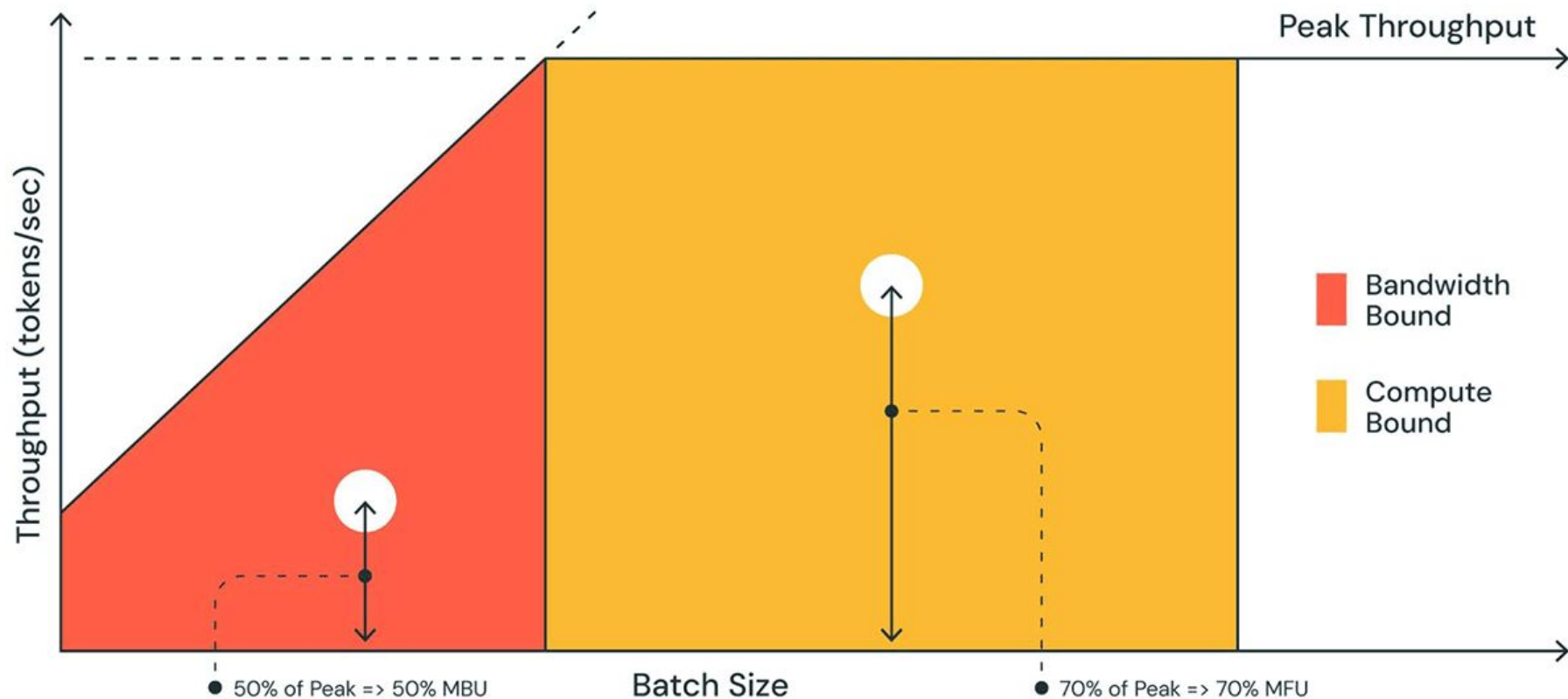
1.58496250072

Хорошо, но мне мама говорила Python медленный!

Just-In-Time Compilation:



Batch vs Throughput



Ячейка памяти на минималках

И как её испортить при исследовании

